

Prelab 4.1 Middleware 1 - MTConnect agent

Learning Goals

Students will be able to:

1. Describe the functionality of MTConnect
2. Implement a connection between computer and MTConnect Agent

1.1 Introduction

Through previous labs so far, we have collected data from sensors (temperature sensor and accelerometer) and IoT devices (power meter-Modbus and vibration sensor-IO-Link). However, someone may raise a question about how to deal with various devices, sensors, and information if each has different data types, sampling frequencies, even communication protocols. To facilitate and easily use various data streams in a single platform, the middleware concept was developed. Middleware is a kind of software that offers common services and capabilities to applications outside of what's provided by the operating system. Data management, application services, messaging, authentication, and API (Application Programming Interface) are all commonly handled by middleware. The main purpose of middleware is to help developers or users build applications more efficiently. There are a lot of middleware in manufacturing industry such as MTConnect, OPCUA, MQTT, ROS, and so on. Each middleware has distinct characteristics, pros, and cons. For example, in the robot industry, [ROS, Robot Operation System](#) is widely utilized as a middleware and open-source platform. In this course, however, we will mainly use MTConnect as a middleware for data aggregation.

MTConnect is an open source and royalty-free standard for manufacturing that is connecting manufacturing equipment with applications by using proven Internet protocols based on XML and HTTP. MTConnect enables real-time data sharing between shop floor equipment such as machine tools and computer systems. In addition, MTConnect offers a semantic vocabulary for manufacturing equipment to provide structured, contextualized data with no proprietary format. With uniform data, developers, integrators, and users can focus on useful, productive manufacturing applications rather than translation (Figure 1). As Figure 1, each machine tool speaks a different language; connectivity and integration are a nightmare without MTConnect standard. Data sources of MTConnect include machine tools, production equipment, PLC, machine controllers, sensors and sensor systems, and other factory hardware.

Applications that consume MTConnect data provide more efficient operations, improved production, optimization, and increased productivity. Therefore, MTConnect stands for: 1) Middleware itself to enable real-time data stream on shop floor and 2) MTConnect standard (ANSI/MTC1.4-2018) which describes a semantic standard vocabulary for manufacturing equipment. In the lab as mentioned above, we will mainly use MTConnect as a middleware for real-time data streaming and applications. If you are interested in MTConnect standard in detail, please visit [MTConnect.org](https://mtconnect.org).



Figure 1 The manufacturing world and applications: before MTConnect (left) and after MTConnect (right)¹

¹Source:

https://mtcup.org/Getting_Started_with_MTConnect_%E2%80%93_Writing_Client_Application/
available on Feb. 6, 2023

1.2 MTConnect Overview

MTConnect as middleware consists of two parts: 1) MTConnect agent and 2) MTConnect adapter as Figure 2. In Lab4 and Lab5, we will explore and practice MTConnect agent and MTConnect adapter, respectively.

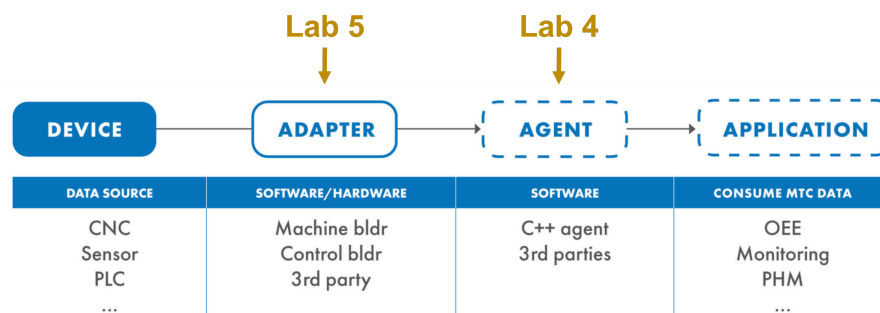


Figure 2 MTConnect configuration and data flow²

MTConnect standard is broken down into four parts. The four standard sections of MTConnect with the major sub-sections are listed below. MTConnect standard documents (version 1.6, released in Sep. 2020) are uploaded on [Github repo here](#). Or,

you can download the standard documents from the MTConnect institute website (<https://www.mtconnect.org/documents>). FYI, the latest version of MTConnect Standard is 2.1 (Jan. 2023).

- MTConnect Standard Part 1
 - Overview and Protocol
 - MTConnect Document Structure
 - Versioning
 - HTTP and XML - Brief Reminder
 - Architecture Overview
 - Request Structure
 - Agent Initialization
 - Application Communication
 - Agent Data and Agent Asset Storage
 - Reply XML Document Structure
 - Devices
 - Streams
 - Assets
 - Error
 - Protocol (Commands to Agent) Overview
 - Probe
 - Sample
 - Current
 - Asset
 - MTConnect Agent and Adapters
- MTConnect Standard Part 2
 - Components
 - Data Items
- MTConnect Standard Part 3
 - Streams
 - Event
 - Samples
 - Conditions
- MTConnect Standard Part 4
 - Assets

²Source: <https://www.mtconnect.org/>, available on Feb. 6, 2023

1.3 GitHub source for MTConnect

Free and open source developer tools are on [Github of MTConnect Institute](#) (Figure 3) and community-maintained. In the MTConnect GitHub page, you will find various

programs, adapters and tools. You can appreciate open-source programs by other developers from GitHub page. All about MTConnect agent are on [C++ agent page](#). If you are interested in release history and descriptions about MTConnect agent, please visit the GitHub page and see README.md.

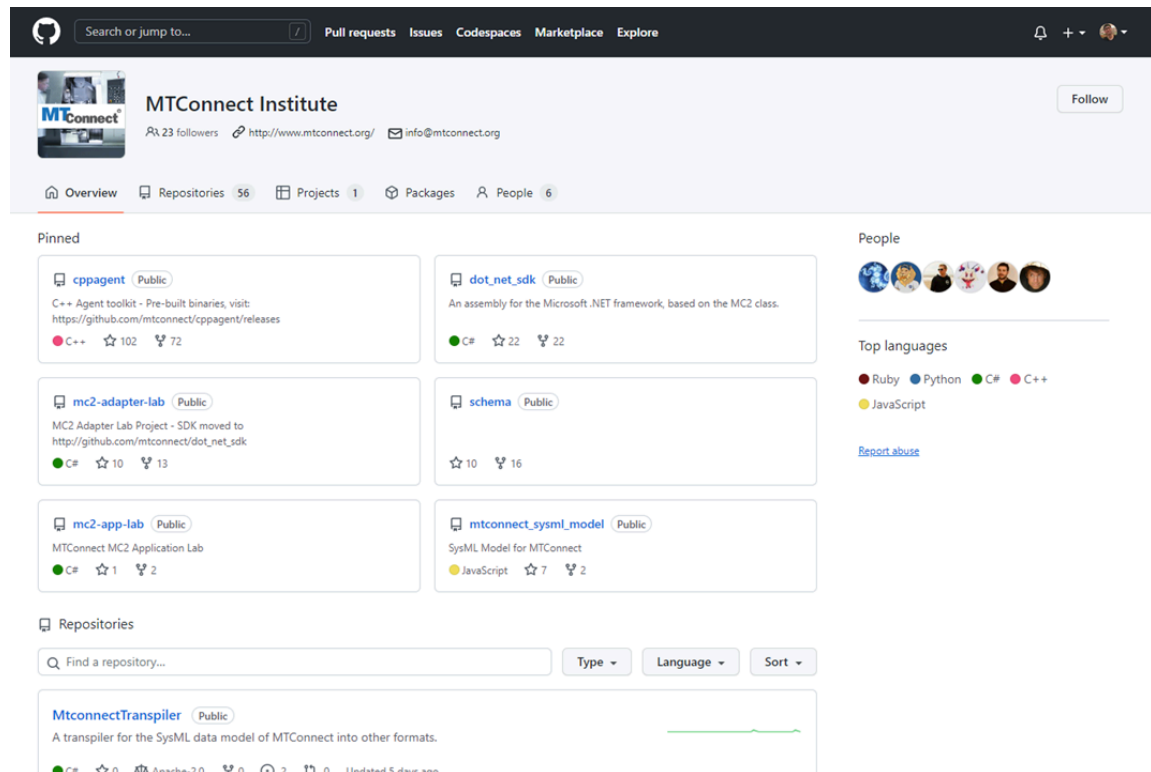


Figure 3 MTConnect GitHub

1.4 Install ruby for simulation

In Lab 4, we will run MTConnect agent on both Raspberry Pi (Linux-based Raspberry Pi OS) and laptop (Windows). All C++ compiled execution files of MTConnect agent on both OS (Operation System) platforms will be given via Github Repo before Lab4. Also, we will simulate MTConnect agent example in Lab 4 based on data we collected in the previous labs. For the simulation, we will use Ruby which is a dynamic, open source, interpreted, high-level programming language. You do not need to know about programming in Ruby. However, it is required to install Ruby on your computers. First, install 'Ruby' on your laptop as followed procedure.

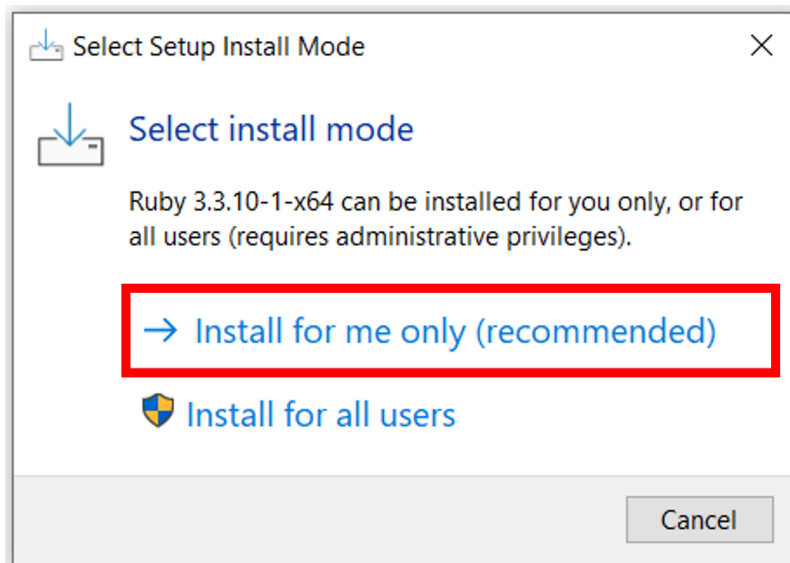
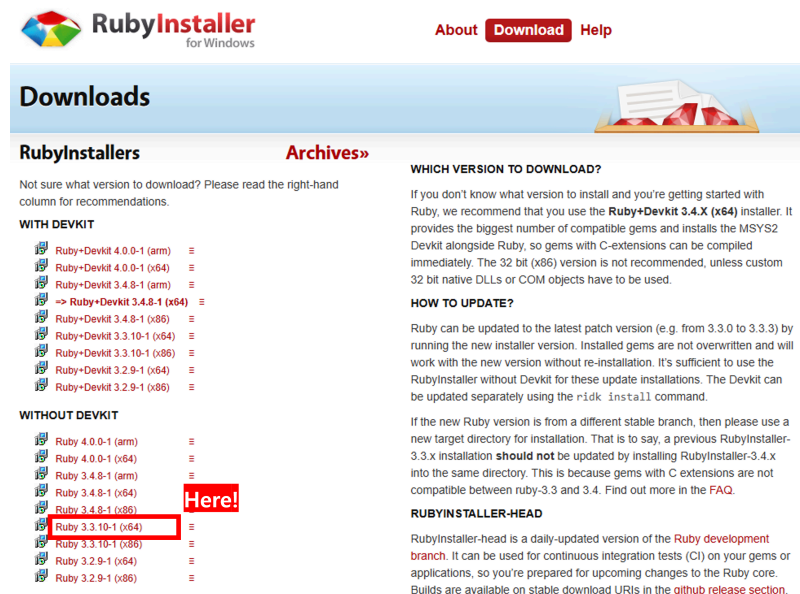
※ If your laptop is Linux, please visit Ruby website via the link below to install it.

- [Ruby installation on Ubuntu using apt](#)

The detailed steps below are for Windows 10 laptop.

1. Download Ruby installer version 3.3.10 without development kit

- a. Download website: <https://rubyinstaller.org/downloads/>
- b. Download 3.3.10-1 (x64) version as shown in the following figure.



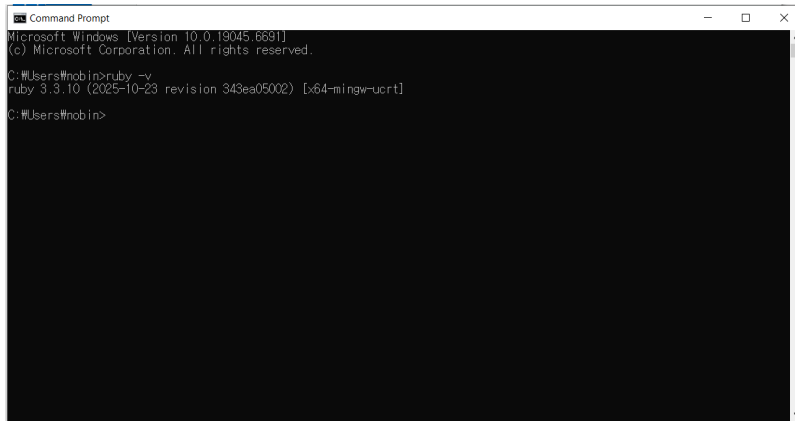
2. Install Ruby by running the downloaded exe. file.
 3. Finish the installation without installing 'MSY2' as below.
- a. In the completing window, uncheck the checkbox and click 'Finish' as below.

```

```

To check the installation of 'ruby' and see the version, use 'Command Prompt' as below.

`ruby -v`



```
Command Prompt
Microsoft Windows [Version 10.0.19045.6691]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nobin>ruby -v
ruby 3.3.10 (2025-10-23 revision 343ea05002) [x64-mingw-ucrt]

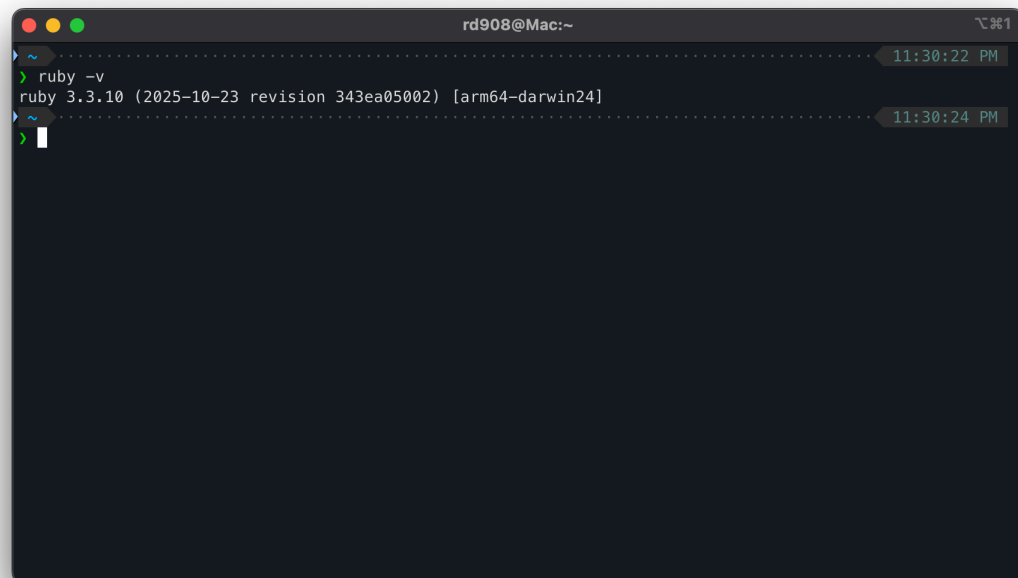
C:\Users\nobin>
```

Figure 4 Ruby version check on Windows

If you get error when checking the version of ruby, try to install 'MSYS2' from <https://www.msys2.org/>

Task 1.1

After installing Ruby on laptop, capture Command Prompt of running 'ruby -v' command as Figure 4 and place on the next block.



```
rd908@Mac: ~
> ruby -v
ruby 3.3.10 (2025-10-23 revision 343ea05002) [arm64-darwin24]

>
```

Second, install 'ruby' on Raspberry Pi as followed procedure.

1. Make Raspberry Pi up to date.

Raspberry Pi - Terminal

```
sudo apt update
```

```
sudo apt upgrade
```

2. Install 'ruby' using 'apt install' command

Raspberry Pi - Terminal

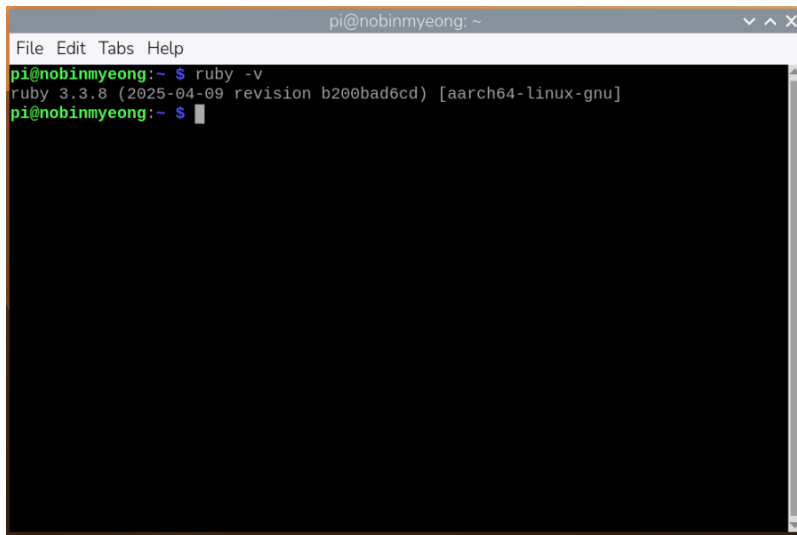
```
sudo apt-get install wine
```

```
sudo apt install ruby -y
```

To check the installation of 'ruby' and see the version, use 'Terminal' as below.

Raspberry Pi - Terminal

```
ruby -v
```

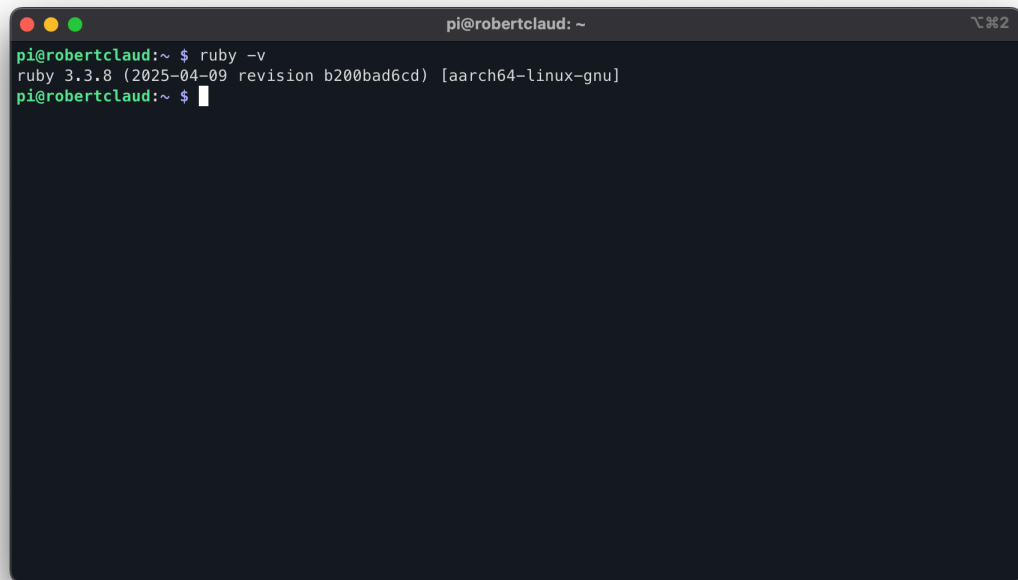


```
pi@noblinmyeong: ~  
File Edit Tabs Help  
pi@noblinmyeong:~ $ ruby -v  
ruby 3.3.8 (2025-04-09 revision b200bad6cd) [aarch64-linux-gnu]  
pi@noblinmyeong:~ $
```

Figure 5 Ruby Version Check on Raspberry Pi

Task 1.2

After installing Ruby on Raspberry Pi, capture Terminal of running 'ruby -v' command as Figure 5 and attach it in the prompt below.

A terminal window titled 'pi@robertclaud: ~' with standard macOS window controls. The terminal shows the command 'ruby -v' being executed, resulting in the output: 'ruby 3.3.8 (2025-04-09 revision b200bad6cd) [aarch64-linux-gnu]'. The prompt 'pi@robertclaud:~ \$' is visible on the next line.

```
pi@robertclaud:~ $ ruby -v
ruby 3.3.8 (2025-04-09 revision b200bad6cd) [aarch64-linux-gnu]
pi@robertclaud:~ $
```

Related Websites and Further Reading

- MTConnect official website: <https://www.mtconnect.org/>
- MTConnect user portal (Community): <https://mtcup.org/>
- MTConnect Institute GitHub: <https://github.com/mtconnect>
- MTConnect C++ agent: <https://github.com/mtconnect/cppagent>
- MTConnect dot net SDK (C#): https://github.com/mtconnect/dot_net_sdk
- MTConnect PocketNC_adapter (Python):
https://github.com/mtconnect/PocketNC_adapter
- MtconnectCore (by TAMS): <https://mtconnectcore.dev/>
- MTConnect agent example (by Mazak): <http://mtconnect.mazakcorp.com/>
- Smart Manufacturing Systems (SMS) Test Bed (by NIST):
<https://www.nist.gov/laboratories/toolsinstruments/smart-manufacturing-systems-sms-test-bed>

Task 1.3

Check the following resources and answer the follow-up questions.

1. Why is middleware needed?
 2. What are MTConnect and its applications?
- a. Summarize MTConnect briefly and possible applications. 3. What are pros and cons of MTConnect?
- a. Compare with other middleware 4. What is XML document? 5. What is HTTP?

```
In [ ]: from IPython.display import YouTubeVideo
        YouTubeVideo('p59jonxxIII')
        #Watch this video.
```

Out[]:



```
In [ ]: from IPython.display import YouTubeVideo
        YouTubeVideo('hUeLAKqWt1M')
        #Watch this video.
```

Out[]:



1) Why is middleware needed?

Middleware is needed so that devices produced by different manufacturers can communicate with one another. Individual devices all have unique communication

protocols, so middleware can be used to translate these unique communication protocols into a common standard languages or protocol.

2) What is MTConnect? Describe some of its applications.

MTConnect is a communication standard that regulates shop floor data from different devices. It acts as middleware to translate data from devices produced by different manufacturers into a single common communication protocol. It uses data dictionaries and semantic data models to describe manufacturing equipment. Some of its applications include machine monitoring, statistical process control, and automation tasks. For example, integrating the status of a robotic arm with a conveyor belt that it drops finished products onto.

```
In [ ]: from IPython.display import YouTubeVideo
        YouTubeVideo('sAityNwwyJE')
        #Watch this video.
```

Out[]:



3) Describe pros and cons of MTConnect. Compare it with other middleware

MTConnect Pros:

- Enables fast deployment of software applications
 - More meaningful contextual data
 - Enables data driven decisions
 - Moves towards a plug-and-play environment
- Cons:

- Can be complex
- Limited control of machines
- Read only so data cannot be sent back to devices.

An example of middleware I have used in the past is Cross-Origin Resource Sharing (CORS) which is particularly useful in web development. Like MTConnect, CORS defines a set protocol for frontend webpages to communicate with a backend or another origin / server. The two middleware differ with their application where MTConnect specializes in factory floor equipment and CORS specializes in web browsing. CORS also differs in that it is not a read only policy while MTConnect is.

```
In [ ]: from IPython.display import YouTubeVideo
        YouTubeVideo('HggTJVtJEag')
        #Watch this video.
```

Out[]:



```
In [ ]: from IPython.display import YouTubeVideo
        YouTubeVideo('_zItx6ndfAs')

        #Watch this video.
```

Out[]:



4) Describe what is an XML document?

An XML document is a document using the Extensible Markup Language that translates machine code into a human readable format. It is very similar to HTML but allows greater flexibility when applying it to factory floor machines since new tags can be created to define different data types.

5) Describe what is HTTP?

HTTP stands for Hyper Text Transfer Protocol. HTTP enables communication of text-based language between clients and servers. It commonly is used in web development but also can be used to communicate with shop floor devices so that information can be viewed in a web browser or application.

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